BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 024 · The Internet, DNS and Cloud Storage

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a NIC provide?

Q2. Valid ages are 13–18. Which is normal test data?
A. 12
B. 19
C. 'twelve'
D. 15

Q3. For a numeric age input, give erroneous data of the wrong type.

Q4. A login asks for a username, password and another factor. “Authentication proves all the user's input is correct.” Correct this. What else needs checking?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Internet — Worldwide system of connected networks.

DNS — Domain Name System; resolves names to IP addresses.

Domain name — Readable name used to identify an Internet service.

IP address — Address used for routing data on networks.

Cloud storage — Remote file storage accessed over a network.

Client — Device requesting a service.

Server — Computer providing a service.

New learning

The Internet: The Internet is a worldwide set of interconnected computer networks where clients can request resources from servers.

DNS: The Domain Name System uses distributed servers to look up IP addresses associated with domain names.

Cloud storage and software: Cloud storage puts files on remote servers; cloud software and processing also provide services using remote computing resources.

Benefits and limitations: Cloud services support access from different devices, but depend on connectivity, service availability and secure access.

Worked example

A pupil enters a website's domain name in a browser. Describe how DNS helps them reach its web server. [4 marks]

1. The pupil begins with a readable domain name, not the server's numeric network address.

2. The device queries a DNS service to find the matching IP address [1].

3. DNS looks up the domain [1] and returns the IP address to the client [1].

4. The browser connects to the web server using that address [1]. Check the sequence.

We Do · Practise with guidance

Guided 1

A visitor enters the name of a local museum website. Describe the DNS lookup and connection in four stages. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: Client asks DNS for an address [1]; DNS finds an IP address linked to the name [1]; returns the IP address [1]; browser connects to the web server [1].

Guided 2

A student stores revision notes on a remote service. Explain one advantage and one disadvantage. [4 marks]

Worked method: Notes can be accessed from different connected devices [1], enabling work at home or school [1]. Internet/provider availability is needed [1], so outages can block access [1].

Guided 3

Explain the difference between a client and a server when watching a web-hosted tutorial. [4 marks]

Worked method: The student's device is a client [1] requesting the tutorial [1]; the remote server hosts/provides the content [1] and responds to the request [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write one line to output the demonstration IP address.

```
domain = "museum.example"
ip = "192.0.2.7"
```

write

Question 1

State one purpose of DNS. [2 marks] Source: Adapted from your Paper 1 revision booklet, 1.3.1.

Question 2

A charity keeps its shared donation records in cloud storage but loses its Internet connection during an event. Explain a benefit and a limitation of cloud storage for this charity. [4 marks] Source: Original J277-style question; not a past-paper original.

Question 3

A visitor enters the domain name of a community sports club's online ticket page. Describe how DNS, the visitor's client device and the web server work together to load the page. [6 marks] Source: Original J277-style question; not a past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

DNS associates human-readable domain names with IP addresses.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The interface that enables a device to connect to a network.

Q2. D — 15

Q3. The text 'twelve' is the wrong data type and must be handled or rejected.

Q4. Authentication checks evidence of a user's identity. It does not prove their input is correct or safe.

I Do / We Do pages · Guided practice

Q1. Client asks DNS for an address [1]; DNS finds an IP address linked to the name [1]; returns the IP address [1]; browser connects to the web server [1].

Q2. Notes can be accessed from different connected devices [1], enabling work at home or school [1]. Internet/provider availability is needed [1], so outages can block access [1].

Q3. The student's device is a client [1] requesting the tutorial [1]; the remote server hosts/provides the content [1] and responds to the request [1].

You Do pages · Code First

```
domain = "museum.example"
ip = "192.0.2.7"
print(ip)
```

Expected result: 192.0.2.7

You Do pages · Independent questions

Q1. It looks up an IP address [1] associated with a requested domain name [1].

Q2. Remote donation records are accessible from different connected devices [1], helping staff use records at events and at headquarters [1]. Cloud access requires an available Internet connection/provider [1], so an outage may prevent the charity from retrieving those records [1].

Q3. Client starts with name [1] and requests DNS lookup [1]; DNS locates address [1] and returns IP [1]; client contacts server [1], which returns requested content

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

[1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: